TOBIAS LOHSE

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Chicago, IL

10+ years turning AI capabilities into production systems, architecting pragmatic solutions and leading nimble teams from ML research to shipping great UX.

9/2023 – Present | 2 years 4 months | Remote

Founding Engineer » Head of Engineering, Orca DB

- Led 5 engineers building an MLOps platform for retrieval-augmented classifiers with millisecond inference and data-centric online learning through memory updates.
- Engineered pipeline for GPU-optimized fine-tuning of embedding models with custom contrastive losses, distributed gradient caching, and automatic batch size tuning.
- Prototyped PyTorch harness for shaping embedding-space decision boundaries (1000x faster feedback), and fine-tuned LLMs for embedding-guided synthetic data generation.
- Implemented Next.js app for statistical analysis, synthetic data generation, and visualization of model context usage, including an AI copilot that can take actions.
- Shipped enterprise-grade multi-tenant auth (SSO, SCIM, audit logs) and drove SOC2 compliance, slashing overhead with CI for vulnerability mitigations and release tracking.
- Built agent-native SDLC platform with eval harness and a compounding skill library.
- Established on-call processes and full-stack observability with distributed OTel tracing.
- Unified release pipeline for cloud (GKE) and on-prem deployments with E2E gates.
- Shipped Postgres-backed background job queue and zero-downtime data migrations.
- Designed the Python SDK and docs to guide developers from quickstart to production.

6/2022 – 7/2023 | 1 year 2 months | Remote

Senior Engineering Manager, Indigo Ag

- Managed team of 7 building tools that paired satellite imagery with soil modeling to target high-potential farmers, driving 80% growth in carbon credit enrollments.
- Coached 2 engineers to promotion, ran book clubs that evolved architecture org-wide, organized 80-person on-site, led full-stack chapter, and improved on-call for 5 teams.

9/2020 – 4/2022 | 1 year 8 months | Remote

Principal Engineer » Associate Director, Tandem Diabetes Care

- Led project to launch a diabetes management platform for 330k patients worldwide.
- Hired 3 managers and 8 engineers, restructured department with 60 contributors into agile teams, and implemented new performance review system across 10 managers.
- Automated generation of regulatory IEC 62304 filings for SaaMD from integration tests.

10/2016 – 10/2020 | 4 years | Potsdam, Germany & Remote

Co-founder & CTO, Cinuru Research

- Built white-label apps with gamified loyalty programs for cinemas including the largest German cinema chain, reaching 200k MAUs and a 4.7 average AppStore rating.
- Implemented newsletter personalization engine that led to 20% ticket sale increases.

1/2016 – 9/2016 | 9 months | Berlin, Germany

Machine Learning Engineer, Aaron AI

- Built NLP call-routing system using TF-IDF, Word2Vec, and random forests.

11/2012 – 1/2015 & 7/2018 – 8/2018 | 2 Years 5 months | Berlin, Germany

Web Developer & Designer, Freelance

- Shipped a React Native dating app in one month and built websites for various clients.

SKILLS

~ 5 YEARS

Machine Learning & AI

Transformers, PyTorch, Scikit Learn, Numpy, Pydantic AI, Vercel AI, Milvus, RAG, Agent Eval, Finetuning, CUDA

~ 9 YEARS

Backend & DevOps

Node.js, Python, REST, GraphQL, RPC, WebSockets, PostgreSQL, SQLite, Redis, OpenTelemetry, Docker, K8s, Terraform, DDD, Event Sourcing, GitHub actions, GCP, Serverless

~ 13 YEARS

Web & App Development

TypeScript, React (Native), Next.js, Tailwind, HTML Emails, Playwright, i18n, a11y, Auth & SSO, Design Systems

EDUCATION

Bachelors supported by Scholarship of Studienstiftung des deutschen Volkes

4/2014 – 12/2015 | FREIE U. BERLIN

Bachelor of Arts in Philosophy & Computer Science

5/2012 – 11/2012 | NORTHWESTERN U.

Bachelor Thesis in Quantum Computing

10/2009 – 5/2012 | U. KONSTANZ

Bachelor of Science in Physics

OPEN SOURCE

OpenAPI HTTPX

CLI for generating typed HTTPX clients from OpenAPI specs

Event Sync

Event sourcing based offline-first state synchronization framework

Styled Native Components

CSS in JS library for React Native